Shailendra Kumar

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Summary

AI/ML and backend engineer with 6.10+ years building scalable, low-latency systems. Currently shipping production GenAI: cross-vertical **agentic AI companions** on the **Model Context Protocol (MCP)**, **LLM** tool-calling with **RAG**, and a Personalisation Engine backed by a **workflow state machine** on **Google Cloud (Bigtable, Pub/Sub)**. Deep experience in Golang, Java, GraphQL, and Kafka, with a proven record of optimizing throughput and reliability across real-time and batch data pipelines.

Technical Skills

AI / GenAI: LLMs (Gemini, GPT), Agentic AI, Model Context Protocol (MCP / FastMCP), LangChain, RAG, A2UI (Generative UI), Prompt Engineering, LLM Observability (Langfuse)
Languages & Frameworks: Golang, Gin, Java, Spring Boot, Python
Databases: MySQL, PostgreSQL, DynamoDB, Redis, MongoDB, Google Bigtable, Cloud Spanner, Apache Doris, NoSQL
Streaming & Messaging: Kafka, Google Pub/Sub, MQTT, Apache Flink, KSQL
Cloud & DevOps: Google Cloud (GKE, Bigtable, Pub/Sub, Spanner), AWS, Docker, Kubernetes, Helm, Git, Airbyte, Prefect
Tools & Technologies: GraphQL, Microservices, Apache Pinot, Apache Parquet, JMeter, Postman, New Relic

Experience

Reliance Intelligence (Jio) Apr 2026 – Present
AI/ML Engineer (Consultant) *Bangalore*

- Architected and shipped a fault-tolerant **Orchestration Engine** using **Redis Streams + Consumer Groups**, delivering **sub-50ms ingress latency** and zero message-loss guarantees for a conversational AI platform.
- Built a production-grade **Multi-Agent Supervisor** with **ReAct loops, parallel tool execution, and guardrails**, orchestrating queries across **9+ AI verticals**.
- Designed a reusable **Python workflow state machine** backed by **Google Bigtable + Pub/Sub**, powering resumable, intent-driven personalization workflows across **9 verticals**.
- Designed a **multimodal AI agent** using **Temporal workflows** to generate personalized home greeting cards and daily news digests based on user preferences, integrating Gemini/GPT routing, TTS, and replay-safe workflow execution.
- Built production **MCP AI companions** integrating **LLM tool-calling, RAG, A2UI, and state tracking** across health, astrology, devotion, and learning verticals.
- Architected **Trust-and-Safety MCP** for fact-checking text/images using **Google Search grounding** and authorized sources to mitigate misinformation.
- Shipped **Finance-MCP** from zero-to-production using **Python/FastMCP**, **Vertex AI Gemini**, Google Search, and **AMFI NAV data**, serving real-time grounded financial answers.
- Built a **multimodal AI + news-digest pipeline** with **Gemini/GPT routing, TTS, replay-safe workflows**, and production observability using **New Relic + GCP**.

Proximity (JioHotstar) Nov 2025 – Apr 2026
Senior Software Engineer (Backend) *Remote*

- Developed, load-tested, and scaled a high-performance **GraphQL backend in Go** powering **40%+ of dynamic sports landing page data** for 500K+ DAU with 99.9% uptime; reduced p95 latency by **45%** and improved throughput by **3x** through parallel resolvers and multi-layer caching.
- Built a distributed **SEO ingestion service** using DynamoDB, Kafka, and CMS/Cricinfo APIs processing **1M+ records/day** with less than 1 minute data lag, improving freshness and reducing manual operations by **90%**.
- Led feature governance using **LaunchDarkly** for tournament-level configurations and progressive rollouts across environments, accelerating releases by **2x** while reducing production risk by **60%**.

Pixis Jul 2023 – Nov 2025
Senior Software Engineer *Bangalore*

- Developed a scalable **Golang** event processing service handling **100K RPM** with a **99% reduction in data loss**. Integrated Kafka for ingestion, implemented retry mechanisms with exponential backoff, and enhanced resilience using **Dead Letter Queues (DLQ)**.
- Designed and implemented a scalable data platform using **Python**, **Golang**, Kafka, S3 Data Lake, and Apache Doris, processing over **1TB of data daily** across **5,000+ pipelines**.
- Developed a workflow orchestration service using **Java Spring Boot** for triggering and monitoring pipelines using event-driven architecture, processing **100K+ requests/day** with **99% success rate**.
- Recovered the **Apache Doris production database** using snapshot-based restoration and Python automation scripts, significantly reducing downtime and improving recovery speed by **98%**.

Threedots Mar 2022 – Jul 2023
Software Engineer *Bangalore*

- Designed and developed an **OTP Login/Signup** flow with mobile verification, reducing fraudulent reward exploitation and lowering associated costs by **25% annually**.
- Built the **Watchlist Service** from scratch, adopted by **50%** of users and improving retention by **15%**.
- Developed a low-latency **Golang Orchestrator Service** handling **80K RPM**, improving downstream security and reducing response time by **20%**.

- Created a **Generic Tag Framework Service** improving data retrieval efficiency by **35%**.
- Built a **User Activity Tracking Service** integrating Mixpanel and WebEngage, increasing actionable analytics metrics by **50%**.
- Contributed to bulk **Notification REST APIs** capable of sending **200M+ push notifications/minute**, improving engagement by **30%**.

Ola Electric Jan 2020 – Mar 2022
Software Engineer *Bangalore*

- Developed a scalable **User Authentication and SSO platform** using JWT-based authentication supporting concurrent logins and multi-organization workflows, handling **50K RPM**.
- Built **Device Authentication Services** using JWT access and refresh token workflows, reducing unauthorized login attempts by **40%**.
- Designed and implemented a **URL Shortening Service** improving usability and increasing adoption by **25%**.
- Developed **Device Whitelist and Management Services** with **95% validation accuracy**, reducing fraud cases by **50%**.
- Contributed to **MQTT-based Communication Services** handling **10M+ messages daily** between vehicles and cloud infrastructure.

Education

Indian Institute of Information Technology, Allahabad Allahabad
Bachelor of Technology in Electronics and Communication *Aug 2016 – Jul 2020*
Tulsi Inter College, Pratapgarh Uttar Pradesh
Intermediate (Science) – 88.4% *2015*
Balbhadra Inter College, Pratapgarh Uttar Pradesh
High School – 80% *2013*

Achievements and Awards

- **The Spotlight Award** – Ola Electric
- **The Warthog of Backend Award** – Threedots
- **Pixis Employee of the Month (EOM)** – Pixis
- **Best Cloud Project (AdTech) Award** at India DevOps Show 2025